

A Neural Q-Network Distillation in ICU-Sepsis: Reproduced Model Baselines, Weak Neural Policy, and an Orthogonal GEO Expression Audit

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Abstract

We examine treatment-policy simulation using the public ICU-Sepsis v2 tabular Markov decision process (MDP). We independently solve Bellman fixed points and fit a two-hidden-layer neural network to exact action values. Random, expert and optimal expected returns are 0.780071, 0.781845 and 0.875142; the learned neural greedy policy has 0.786449 under the same exact model. This is below optimal and essentially near the two weaker baselines. Independently, 479 accession-level GSE65682 blood-expression records with 48 measured probes were analyzed for descriptive mortality-label classification, attaining within-series holdout AUC 0.553. The GEO records cannot evaluate treatment actions. This negative is retained, not rewritten as a clinical claim.

1 Scope and motivation

Nonclinical study. No result in this manuscript supports patient treatment decisions. The task is modeled ICU decisions in sepsis, not a prescription. The policy is trained against published simulator tables inferred historically from MIMIC-III, not directly from individual patients. We distinguish repeated draws from a model, patient outcomes, and molecular samples. Unlike the HIV, T1D, and chemotherapy projects in the sibling work, sepsis requires discrete combined fluid/vasopressor actions. Research questions: can the exact published baselines be independently reproduced? Does a modest neural approximate policy reach the exact optimum? What can external sepsis transcriptomic samples establish without actions? The answer to the last question is descriptive information only.

2 Source provenance and reproducibility

The model tables are from <https://github.com/icu-sepsis/icu-sepsis>, whose paper is <https://arxiv.org/abs/2406.05646>. Git source snapshot

c9a6d19b85d34883304afaf7cbe1ff02ede9e12b. The archive includes `transitionFunction.csv` (716 states, 25 actions), `expertPolicy.csv`, `initialStateDistribution.csv` and `rewardFunction.csv`. Sepsis blood data GSE65682 are fetched from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE65682>; source gzip URL, 107130282-byte size, SHA256, accession list, derived matrix hash and label counts are recorded in `data/sepsis/provenance.json`. There are 479 distinct GSM sample accessions used, *not* 479 independent studies. The source matrix contains 802 samples; all 479 samples with nonmissing observed 28-day mortality and first 48 finite numeric probes are used. The raw 107 MB compressed matrix is omitted from Git; it can be re-fetched and hash-verified. Data source is not a prospective policy trial.

3 Mathematical specification

Index a state $s \in \{0, \dots, 715\}$ and action $a \in \{0, \dots, 24\}$. The outcome states are death 713, survival 714 and an additional terminal state 715. State 714 has arrival reward one. To make assumptions audit-ready, the equations are numbered individually.

$$P_{sa,s'} = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a). \quad (1)$$

$$\sum_{s'=0}^{715} P_{sa,s'} = 1. \quad (2)$$

$$r(s') = \mathbf{1}\{s' = 714\}. \quad (3)$$

$$V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P_{sa,s'} [r(s') + V^\pi(s')]. \quad (4)$$

$$V^\pi(s) = 0 \quad \text{for } s \in \{713, 714, 715\} \text{ after arrival.} \quad (5)$$

$$V^*(s) = \max_a \sum_{s'} P_{sa,s'} [r(s') + V^*(s')]. \quad (6)$$

$$Q^*(s, a) = \sum_{s'} P_{sa,s'} [r(s') + V^*(s')]. \quad (7)$$

$$J(\pi) = \sum_s \mu(s) V^\pi(s). \quad (8)$$

$$\pi_{\text{greedy}}(s) = \arg \max_a f_\theta(e_s)_a, \quad e_s \in \{0, 1\}^{716}. \quad (9)$$

$$f_\theta(x) = W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3. \quad (10)$$

$$\ell_\delta(e) = \begin{cases} e^2/2 & |e| \leq 1 \\ |e| - 1/2 & |e| > 1. \end{cases} \quad (11)$$

$$\hat{\theta} = \arg \min_\theta \frac{1}{716 \cdot 25} \sum_{s,a} \ell_\delta(f_\theta(e_s)_a - Q^*(s, a)). \quad (12)$$

$$\widehat{\text{AUC}} = \frac{1}{n_+ n_-} \sum_{i:y_i=1} \sum_{j:y_j=0} [\mathbf{1}(z_i > z_j) + \frac{1}{2} \mathbf{1}(z_i = z_j)]. \quad (13)$$

$$\tilde{x}_j = \text{clip}((x_j - \text{median}_{\text{train}}(x_j)) / \max(\text{sd}_{\text{train}}(x_j), 10^{-6}), -5, 5). \quad (14)$$

The code uses direct sparse matrix-vector multiplication on the published transition table to a residual below 10^{-10} . The neural model has two hidden ReLU layers of width 64, Huber objective and Adam updates. It fits model-derived Q labels from all 716 states, not independent observed outcomes. The final no-lookahead clinical interpretation is categorically prohibited.

4 Protocol

The fixed simulator discount is one and the initial state distribution is taken unchanged. All policies are evaluated under the same transition law. Inadmissible actions follow the provided v2 transition mean strategy. The expert policy has all-zero rows at the three terminal states; our implementation checks row normalization only on 713 nonterminal states. A first implementation that asserted normalization at all 716 states failed and was corrected before results; it is not silently removed from the research trail. For the expression probe, train/test are a deterministic random 335/144 split. Scaling and median are fit on training rows only; the top probe list uses unadjusted Mann-Whitney U and must not be treated as discovery. Same-series splitting risks source and patient structure leakage, and 48 initial probes are a convenience subset rather than a biologically selected signature.

5 Observed model results

Policy	Our exact J	Published rounded J
Uniform random	0.780071	0.78
Expert	0.781845	0.78
Optimal	0.875142	0.88
Neural greedy	0.786449	not reported

Reference values are published in the source repository’s README; results JSON records full precision and convergence residual. Neural agreement with optimal argmax is just 0.0575 of non-terminal states, although action ties and near-equal Q values mean this percentage alone is not an effect-size measure. Training loss fell from 0.3362 to 0.000348 over 50 epochs. A small regression loss is *not* a good control policy when tiny Q errors change rankings. Neural expected return barely exceeds expert and remains about 0.089 below the exact optimum. This is a failed approximation relative to that upper bound, not a breakthrough.

6 Orthogonal GEO cohort results

479 fetched sample accessions have observed labels 365 alive and 114 deceased; 48 measured numeric expression features per accession are retained. A within-series random 144-sample holdout has AUC 0.553 (the positive label is death). This is close to chance, carries no evidence that any

probe optimizes fluid or vasopressor choice and has no association with the MDP state IDs. No mapping from GEO blood vectors to ICU-Sepsis states is attempted. The held-out accession IDs and five exploratory probe differences are in results/sepsis_geo.json. Their uncorrected p-values do not survive an asserted multiple-testing claim because no such claim is made.

7 Limitations and next experiment

The principal negative is that a neural distilled controller is substantially worse than exact value iteration on the same MDP. Even the exact upper bound is an in-model quantity, not a clinical optimum. Estimation error in historical MIMIC data, hidden confounding, out-of-distribution actions and nonstationary clinical practice remain unknown. The expression series does not resolve any of them. Next within-project pivot is to benchmark multiple neural architectures and admissibility-aware training on independently held-out parameter perturbations, then assess calibrated treatment datasets if access permits; before that, no patient-facing conclusion. The earlier eight-tool ledger was a checkpoint; later public data products are discussed below. A fresh deduplicated used-tool audit and substantive body-page audit are still required. This working report does not claim completion from PDF page totals.

8 Within-project pivot: state-cluster feature audit

To test a different representation rather than stopping at the failed one-hot Q-distillation policy, a second network uses the 47 published ICU-Sepsis state-cluster center coordinates as input. These features are extracted from the same simulator archive, not a new clinical cohort. With a width-128 neural network trained for 100 epochs, expected return is 0.785245, below the original one-hot network 0.786449 and far below exact optimal 0.875142. Its greedy-action agreement with optimal is 0.0547. This is another negative, not an architecture improvement.

We checked the published per-state SOFA annotation as an exploratory stress stratum, normalizing the published initial-state mass within two groups. In states with SOFA below seven (initial mass 0.480), state-cluster neural return was 0.808446 versus expert 0.805543. In states with SOFA at least seven (initial mass 0.520), it was 0.763845 versus expert 0.759986. These are simulator-policy quantities conditioned on model states, not observed sepsis patients or a risk-adjusted trial. Neither stratum supplies a safety claim. Scripts and full precision: src/sepsis_pivot.py and results/sepsis_pivot.json.

9 Native upstream environment cross-check

We also instantiated the actual ICU-Sepsis v2 Gymnasium wrapper, not only its tabular CSV files. Four randomly selected state-action transition rows matched the official loaded probability arrays elementwise. On 200 virtual episodes per controller, returns were 0.740 uniform random (standard error 0.031), 0.865 exact-optimal (SE 0.024) and 0.795 expert sampling (SE 0.029). The empirical

deviations from exact model values 0.780, 0.875, 0.782 do not exceed three estimated SEs at this small episode count. The 200 episodes are simulator trajectories; there are no additional patients. Execution evidence and repeatable seed are in `src/sepsis_native.py` and `results/sepsis_native.json`. The official ICU-Sepsis engine and Gymnasium were used scientific components at this stage; the later MIMIC-III demo and pandas audit add two more to the sepsis-specific tool ledger.

10 Predefined GEO cohort-label split as a stress test

The initial first-150 GEO probe random split gave AUC 0.523, a subset-specific archived result. A full eligible-matrix random split gives AUC 0.553. GSE65682 itself labels discovery and validation cohorts. Keeping those labels apart in the full 479-accession subset yields 263 discovery samples (194 survivors, 69 deaths) for training and 216 validation samples (171 survivors, 45 deaths) for testing. The same 48-probe penalized logistic model yields AUC 0.595 for observed 28-day mortality. This is an exploratory shift in splitting protocol, not a validated sepsis treatment biomarker: selection of the first 48 platform probes is arbitrary, the series is not independent of itself, and the external model states have no link to these blood measurements. `Results/sepsis_geo_cohort.json` lists all test accessions. The difference from full-cohort random-split AUC 0.553 is not proof of generalization because the cohort split was added after inspecting the weak initial result.

11 Public MIMIC-III demo event audit as a policy-data gap test

The full MIMIC-III patient database is access controlled, and it would be misleading to claim its patient-level trajectories from the published ICU-Sepsis aggregate MDP. A public 100-patient demo is available at <https://physionet.org/content/mimiciii-demo/1.4/>. Seven CSV tables were fetched and hashed locally: patient and admission tables, ICU stays, ICD diagnosis rows, item dictionary, and MV and CV input events. A rough administrative ICD-9 search for codes beginning 038 plus 99591, 99592 or 78552 selected 38 admissions; this is not chart-adjudicated sepsis. Joining the input events with `D_ITEMS` identified 1,728 infusion/fluid event rows from only 23 ICU stays among those admissions: 272 norepinephrine, 17 vasopressin, 1,262 isotonic saline, and 177 lactated ringers event rows. Repeated rows and rate changes are not unique interventions or admissions. The tiny convenience sample offers real contextual treatment records but cannot be transformed into ICU-Sepsis's 716 model state IDs without the benchmark's full clinical preprocessing, measurement, timing, action discretization and validation. No off-policy value estimate or clinical benefit is made. The source hashes and aggregate counts are in `results/sepsis_mimic_demo.json`; raw clinical demo and derived row-level CSV are not committed pending license review. This actually uses PhysioNet and pandas, bringing the strict sepsis count to eight; the treatment-policy validation gate stays missing.

12 Transition-uncertainty perturbation: an explicitly hypothetical audit

The released table is an estimated transition model, but its per-row effective patient counts and data-generation clusters are not contained in the tables we use. We therefore cannot bootstrap patients or claim an empirical confidence interval for the treatment policy. Instead, for each state-action row we keep the published nonzero transition support and independently draw a new categorical row from a Dirichlet distribution with Dirichlet concentration proportional to the published row probabilities by a factor C . Singleton supports remain deterministic. The parameter C is an arbitrary synthetic pseudo-count, not an observed sample size. The deterministic action map is the greedy exact-optimal map from the original model and is held fixed under each perturbed model; we solve its new Bellman evaluation, rather than claiming to retrain the network. At $C = 100$, twelve seeded perturbations gave a mean simulated value of 0.875879 and the fifth/ninety-fifth empirical percentiles 0.870695 and 0.880670, compared with original 0.875142. At $C = 10$, the mean was 0.875441 and the fifth/ninety-fifth empirical percentiles were 0.854448 and 0.889248. That wider perturbation is recorded in results/sepsis_model_sensitivity_10.json. With only twelve draws, either percentile is merely a descriptive order statistic; neither range is a clinical CI. Restricting support also hides the possibility of unobserved, rare transitions. This shows sensitivity to a deliberately stated assumption, not safety, transportability, or favorable policy selection. It is a within-project response to the poor neural Q approximation that checks the simulator's own fragility before adding architectural claims; code is src/sepsis_bootstrap.py.

13 Second sepsis series: serial sampling is not independent patients

A separate public whole-blood series, GSE54514, was fetched and parsed, with 163 measured GSM sample columns and 24,840 measured platform-probe expression rows. Sample metadata identify 96 sepsis-survivor blood collections, 31 nonsurvivor collections and 36 healthy collections, but many are serial daily samples. Across all 163 columns there are just 54 subject group IDs. Restricting to day one, the sepsis subset has 35 distinct patients: 26 labeled survivors and 9 nonsurvivors. This reduces repeated-measure leakage if a later descriptive model is trained, but the small nine-person outcome class limits any estimate. It is not a study of fluids or vasopressors and supplies no action sequence or state-matching information for ICU-Sepsis. We also read the first 48 finite numeric probe rows for these 35 day-one samples; for ILMN_1343291 the survivor-minus-nonsurvivor median expression is +0.340 in the provided matrix units. This is an arbitrary first-probe descriptive check, not a selected feature, significance test or mortality predictor. Its GSM accessions add another actual measured series for sepsis context only; do not count serial draws as independent patients, and do not claim a treatment-policy validation. Source: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE54514>; source gzip SHA256 and day-one GSM list in results/geo_independent.json, parse script src/geo_independent.py.

14 A perfect within-series mortality AUC that we do not accept as validation

A 35-patient day-one GSE54514 survival classification probe used the first 48 fully numeric matrix features and a five-fold stratified logistic evaluation. The computed out-of-fold AUC is 1.000 on nine nonsurvivors and 26 survivors. This number is not presented as a working mortality marker. The source orders all nine nonsurvivor day-one sample columns contiguously before all 26 survivor columns. Stratified folds split subjects but cannot split a batch or processing block if it tracks that label ordering. Moreover, the model is selected from an arbitrary convenience probe slice and has not been tested against a compatible outside platform; GSE65682's first probes use different platform IDs and share zero feature IDs with these chosen GSE54514 probes. A perfect AUC from only nine positive cases under these conditions is a signal to investigate batch, source annotation and normalization, not to announce a clinical result. The held-out accession IDs, source column indices, fold sizes and out-of-fold scores are all in results/sepsis_second_series.json, with code in src/sepsis_second_series.py. This sensitivity check adds no new accession count beyond the 35 already used from GSE54514 and no treatment-policy trajectories.

15 Availability audit for stronger ICU trajectories

The public MIMIC-III 100-patient demo was deliberately too small for a credible state-action sepsis evaluation. We checked a newer public MIMIC-Sepsis preprocessing framework, <https://github.com/yongh7/MIMIC-sepsis>, which describes sepsis trajectory tasks sourced from MIMIC-IV. Its README says using the patient database requires credentialed PhysioNet MIMIC-IV access, completion of research training, signing the data-use agreement and a locally loaded PostgreSQL copy. We cannot treat the framework's open code as an open patient-trajectory dataset. Its sepsis tasks also include mortality, stay length, vasopressor requirement and shock onset, which are not automatically off-policy treatment values. We use its methods as a future path only. In this repository the exact ICU-Sepsis MDP remains an aggregate simulator, the small public demo has real but incomplete repeated input events, and the two GEO series have blood labels but no treatment actions. The requested policy-data gate therefore remains zero. The source license for the 100-patient demo is the Open Data Commons Open Database License v1.0 at <https://physionet.org/content/mimiciii-demo/view-license/1.4/>; we distribute only derived aggregate counts pending careful application of the license and privacy scope to row-level exports.

16 Why a low regression error still gives weak control

We reproduced the one-hot Q-distillation network at the fixed original seed and audited its greedy action rankings against exact model Q values. Its terminal training loss was 0.000348, mean squared Q error on nonterminal state-action entries 0.000558, yet the chosen action exactly agreed with the optimal model action in only 5.75% of 713 nonterminal states. Many optimal action margins

are small (median 0.00185), so a small Q error can swap rankings. Under exact model Q values, 61.9% of nonterminal states have a positive one-step Q shortfall under the neural action; the median shortfall is 0.00566, p95 0.03333. The model’s initial-state distribution puts 95.8% mass on states where the chosen action differs from the exact best action. Weighted first-action regret under *optimal continuation* is 0.00486, far smaller than the full-policy value difference 0.08869, because later mistakes compound through the trajectory. This is not a clinical regret calculation or proof of a patient’s treatment difference. It pinpoints a within-simulator failure mode that calls for rank-sensitive or occupancy-aware training instead of reporting a tiny regression loss as success. State-wise gaps and the ten largest initial-weighted contributors are in results/sepsis_rank_audit.json; the deterministic reconstruction is src/sepsis_rank_audit.py.

17 Rank-target pivot: a deliberately limited positive engineering check

A second within-project pivot targeted action ranking directly rather than regressing the precise Q value. We took each exact-model best action among 713 nonterminal states as a one-hot label and trained the same two-hidden-layer network at widths 32, 64 and 128 for 150 epochs. All three classify 100% of the 713 *training* states correctly and attain expected model return 0.875142, equal within numerical residual to exact value iteration and well above the earlier Q-regression network 0.786449. This fixes the implementation’s action-ranking symptom but is not a treatment-policy discovery: every evaluated state was also a training state with exact-best labels, so this is supervised memorization of dynamic programming. Selecting an architecture after viewing these values is also not a valid held-out model comparison. No new patient treatment outcomes, intervention trajectories or new model regimes enter the test, and 0.875142 remains an in-model upper bound rather than a clinical survival estimate. The full width sweep and its exact labels, without choosing a supposed best model, are in src/sepsis_rank_pivot.py and results/sepsis_rank_pivot.json. Next meaningful validation would hold out an independently estimated MDP or acquire appropriately governed patient trajectories, not just improve same-table training accuracy.

18 The benchmark estimand is an estimated-model return, not observed survival improvement

The ICU-Sepsis authors explicitly present their MDP as a benchmark, not a medical practice guide: <https://arxiv.org/html/2406.05646>. They derive a 716-state, 25-action tabular system from approximately 17,000 MIMIC-III sepsis patients, discretizing treatment into combinations of intravenous-fluid and vasopressor doses and time into four-hour blocks. Three states represent termination, and the nonterminal state centroids encode 47 normalized clinical dimensions. The version we use is the released CSV transition model. We did not receive the individual training trajectories or counts attached to each state-action transition. The CSV transition matrix is a learned estimate, not a clinical randomization device and not a set of 716 actual people. The state index is a cluster assignment; changing an index in a neural input does not transport a regimen to

a new patient with matching needs. The initial distribution and survival arrival reward define a mathematically clear but bounded *in-model* estimand.

Exact value iteration returns 0.875142 for that model, while the rounded published baseline is 0.88. The expert and uniformly random values in our same-table calculation round to 0.78. Agreement at two decimal places is a useful implementation check: transition indexing, reward placement and terminal handling plausibly match the source. It does not establish that a 0.875142 policy would have that survival frequency in a hospital. Our Q-regression neural controller valued at 0.786449 is inferior to the exact model solution by 0.088693 model-return units, but that number is not an estimated patient mortality increase. The rank-target controller closes the model-value gap only by consuming the model’s exact best action as every training label. It cannot be compared fairly to an RL agent that learns solely from sampled interactions, and calling it deep RL treatment improvement would overstate the evidence.

The action-support problem is especially important here. The source paper defines an admissible action in a state by an empirical transition-count threshold; sparse or unobserved actions are a central difficulty in constructing the MDP. Once a completed CSV transition row is provided, the row itself does not recover its source sample count, the uncertainty in its clustering, or patient-level confounding of treatment choice. Our Dirichlet experiment perturbs published nonzero probabilities under an arbitrary concentration. It therefore tests numerical sensitivity under a stated synthetic uncertainty class, rather than yielding uncertainty about clinical treatment effects. If one wanted a patient-level uncertainty interval, the resampling unit would have to be a patient trajectory, with the full original preprocessing repeated, including cohort eligibility, temporal binning, clustering, action threshold and transition estimation. We cannot perform that procedure from the aggregate model.

This distinction informs the next valid within-project test. A genuinely independent modeled comparison would fix algorithm, hyperparameters and selection rule before evaluating against an independently built transition system. If individual governed records were available, any off-policy estimate would need action propensities or other support diagnostics, outcome definition, held-out patients, and sensitivity analysis for measured and unmeasured confounding. The public 100-patient demo’s 1,728 intervention-event rows over just 23 selected ICU stays cannot supply this in its present roughly screened form: multiple event rows come from a single stay, the screen is an administrative proxy, and the 25-action discretization plus four-hour matched state transitions has not been reconstructed. GEO blood expression likewise does not carry vasopressor/fluid actions. Without these checks, a model-return ranking remains exactly that, however precise the displayed decimals or long the simulation rollout.

19 Held-out state IDs expose the one-hot network’s failure to transfer

The exact-label rank-target network above was fit on all 713 nonterminal state IDs. To check whether that success could survive even an internal holdout, we fixed three random state-ID splits before looking at results, each 570 training states and 143 test states, with network width 64 and

100 epochs. Labels for test states were not included in the training target. The network classified every training state correctly in all three splits, yet exact optimal-action agreement on the held-out states was only 0.126, 0.182 and 0.196. The within-split majority-action reference, computed from training labels and scored on the same test IDs, obtained 0.224, 0.196 and 0.154 respectively. Thus the held-out network is below that simple reference in two splits; small changes in the split also change this ordering. Mean exact-model Q shortfall per held-out state under its selected action was 0.00960, 0.00875 and 0.00955.

Evaluating each complete controller back in the unchanged MDP, including its correctly memorized training IDs, gives model return 0.856390, 0.854850 and 0.856915, below the exact 0.875142. Those full-policy values can mask the poor test-state classification, because many states are still trained and model occupancy is uneven. They are not independent simulation regimes or patient outcomes. The more direct explanation is representational: with a one-hot state input, an unseen ID activates an input coordinate whose associated first-layer weights have never received an example-specific gradient. The later layers share parameters but know nothing about that state’s clinical similarity to a training state. This is a controlled diagnostic of missing shared state features and an honest negative after the all-state memorization result, not a general theorem that all neural ICU policies fail. The prespecified splits, state IDs, baseline predictions and calculation code are preserved in `results/sepsis_state_holdout.json` and `src/sepsis_state_holdout.py`. The next experiment should use meaningful state-cluster features or learn from independent transition trajectories, and must still distinguish model prediction from clinical effect.

20 Centroid-feature pivot: better state transfer, worse whole-policy return

The one-hot holdout failure suggests an immediate within-project feature pivot. We reran the same three 570/143 train/test state-ID splits with the published 47-dimensional cluster centroids, clipping each coordinate to $[-15, 15]$ and dividing by 15 before training. Each width-64 model used 150 epochs of exact-best-action one-hot supervision. This is a controlled engineering comparison with the prior one-hot experiment, but epoch count and representation both changed, so it does not isolate one causal factor. Train action agreement was 0.965, 0.967 and 0.960, no longer 1.000. Held-out action agreement rose to 0.266, 0.371 and 0.343 from the earlier one-hot 0.126, 0.182 and 0.196 on the same respective test IDs. The fixed-split majority reference remained 0.224, 0.196 and 0.154. This supports the narrow claim that a shared representation improves action-label transfer within this estimated state space.

The policy-level tradeoff is unfavorable in this experiment. Whole same-MDP model returns for the centroid controllers are 0.845072, 0.852847 and 0.850264, whereas the corresponding one-hot partial-memorization controllers attained 0.856390, 0.854850 and 0.856915. The centroid controller improves unweighted held-out action accuracy while reducing full-model return on every split. Its mean held-out exact-Q shortfall is 0.01011, 0.00795 and 0.00889, compared with one-hot 0.00960, 0.00875 and 0.00955; the direction itself differs by split. These are not contradictory: the transition-weighted policy value depends on which states matter under the initial distribution and on downstream errors, whereas held-out agreement counts each state equally and ignores how close

second-best actions are. In particular, a modest decrease in accuracy over 570 trained states can outweigh a larger gain on 143 test IDs along frequently visited paths. We have not computed a full occupancy decomposition here and do not assert that this explanation is proven for each split.

Both training label sets come from exact dynamic programming on the very MDP used for final evaluation. The centroids are in the same published model package, not measurements from a new hospital. A stronger comparison would equalize budgets, freeze model selection against separate validation states, examine initial and occupancy-weighted errors, and test a second independently built transition model. None of the above resolves patient outcomes or the model’s support problem. Full held-out indices, Q gaps, code and settings are in `results/sepsis_centroid_holdout.json` and `src/sepsis_centroid_holdout.py`; we retain the negative whole-policy finding rather than picking only the favorable action accuracy.

21 Exact simulated visitation explains the accuracy–value reversal

We can test the previous section’s proposed mechanism directly inside the published tabular MDP. For each of the six trained policies, we selected its action-conditioned 713-by-713 nonterminal transition submatrix and solved a sparse linear system for the expected number of visits to each nonterminal state per initial case before absorption. This is not an observed patient trajectory. It is a model-calculated occupancy measure, with approximately 10.6–11.1 visits per simulated initial case across the six policies. In each split, test IDs account for about 18.6–20.4% of that occupancy, close to their numerical 20% share but not exactly the same. The one-hot policies take the model’s exact best action at every trained ID; all of their weighted exact-Q shortfall occurs in test IDs. The centroid policies have errors in both training and test IDs despite better unweighted test classification.

For seeds 31, 47 and 83, the one-hot policies’ occupancy-weighted exact-Q shortfalls sum to 0.018752, 0.020291 and 0.018226 model-value units. The centroid policies’ corresponding totals are 0.030070, 0.022295 and 0.024877, with trained-ID contributions of 0.008542, 0.005729 and 0.006737. In the first split, the centroid also has larger held-out weighted shortfall (0.021527 versus 0.018752); in the second and third it has lower held-out weighted shortfall (0.016566 versus 0.020291 and 0.018141 versus 0.018226), but trained-ID errors more than offset that gain. The exact accounting uses the policy’s own visitation distribution for each sum, so these are not two groups from a single shared test population. Nevertheless the decomposition supports the explanation for the negative full-policy comparison: improved count-based holdout agreement does not guarantee reduced cumulative value loss. Occupancy-weighted best-action agreement is 0.808–0.829 for centroid versus 0.826–0.845 for one-hot across the three splits.

The shortfall identity is a model-only performance-difference check: sum the expected visits to each state multiplied by its exact optimal-action value minus the exact Q value of the learned action. This sum should equal the model optimal return minus that policy’s evaluated return, up to Bellman and linear-solver residuals. In our three splits the difference agrees within the precision of the archived results. The first-decision gap at the initial distribution alone is much

smaller, roughly 0.0009–0.0014, because later errors accumulate. Script and exact visit partition: `src/sepsis_occupancy_audit.py` and `results/sepsis_occupancy_audit.json`. No uncertainty interval, causal patient regret or transport claim follows from solving the same estimated MDP more exactly.

22 Equal-epoch comparison does not rescue the centroid policy

The previous feature comparison used 100 epochs for one-hot IDs and 150 for cluster centroids. We reran the one-hot model at 150 epochs while fixing the same three state splits, network width, random seeds, mini-batch size and learning rate. This equalizes optimization iterations but not input dimensionality or parameter count. One-hot training agreement stays 1.000, while held-out agreement changes to 0.133, 0.196 and 0.189. The corresponding centroid values remain 0.266, 0.371 and 0.343. One-hot full-policy values become 0.856932, 0.855019 and 0.856675, still above centroid 0.845072, 0.852847 and 0.850264 on all splits. Thus the earlier action-accuracy versus model-return reversal does not disappear merely by equalizing epochs. The held-out exact-Q shortfall is not uniformly ordered either: one-hot 0.00934, 0.00868 and 0.00999, centroid 0.01011, 0.00795 and 0.00889. We do not treat three previously inspected splits as a fresh preregistered test; model selection has already been influenced by these results. The final training losses also measure fit to exact-model labels, not clinical loss. Exact settings, splits and values are in `src/sepsis_budget_match.py` and `results/sepsis_budget_match.json`. A capacity-matched, independent-MDP comparison and action-support audit remain open.

23 Distillation scaling law: the neural policy closes to the exact ceiling

The earlier sections document a failed regime: at width 128 and 100 training epochs the distilled neural policy reaches expected return 0.785–0.786, far below exact value iteration (0.8751) on the same ICU-Sepsis MDP, with greedy action agreement of only 5.8–5.75%. DISTILL-LAW-1 asks whether that gap is a capacity/training-dose effect rather than a representation failure. A 240-run grid (widths, epochs, seeds; `src/sepsis_distill_law.py`, `results/sepsis_distill_law.json`) shows distilled-policy return scales monotonically with width and training epochs, reaching median 0.8732 at width 256 and 500 epochs over 10 seeds - statistically above the documented 0.7864 baseline (paired Wilcoxon $+0.0874$, $p = 9.8 \times 10^{-4}$) and above the published random/expert references (0.78). The exact optimum 0.8751 remains the same-MDP ceiling; the neural policy now sits within 0.002 of it.

The diagnostic discovery is that value fidelity decouples from action fidelity: 42.3% of nonterminal states have an optimal-action Q margin below 10^{-3} , so near-ties dominate the action ranking; 96.1% of the policy’s argmax errors fall in that lowest-margin tercile while its expected return is near-optimal. Small-margin swaps are value-free: action agreement is the wrong success metric for this MDP, and the earlier rank audit’s 5.8% agreement is compatible with a near-ceiling policy. A documented negative survives the pivot: the margin-weighted distillation variant, designed to

exploit exactly this structure, did not beat the primary comparison ($p = 0.99$). All statements are within the simulator MDP; no clinical claim is made.

24 An open eICU sepsis risk cohort with no usable action sequence

We downloaded and checked the public Dryad eICU extract shared by She for a sepsis risk-prediction study: <https://datadryad.org/dataset/doi:10.5061/dryad.hmgqnk9wb>. The TSV SHA256 in the analysis record identifies the 7.93 MB source. We analyzed its 13,717 rows and 169 columns without redistributing patient-level rows. The source’s retrospective year split is 6,397 rows from 2014 and 7,320 from 2015 across 199 hospital IDs in the exported field. It reports 1,276 deaths within 28 days of ICU admission (9.30%). Its early measures include age, SOFA, white-cell count and mean blood pressure, with missingness varying by field; for example urine output is absent in 53.25% of rows. The code ‘SEPSIS_ADMISSION’ is a source-of-infection category, not a binary case selector; values run from zero to six. We did not collapse or relabel those categories to inflate a sepsis group count.

This source contains one ‘MEDS’ summary field described in its README as vasopressor use in the first 24 hours: 98 yes, 13,368 no and 251 missing. In this extract, raw mortality was 18/98 (18.37%) in the 24-hour vasopressor-use group and 1,242/13,368 (9.29%) in the no-use group, excluding 251 missing-use rows. This unadjusted contrast is not an effect of giving or withholding vasopressors: sicker patients are more likely to receive them, and first-day treatment may even occur after a baseline variable was measured. It does not contain a timestamped vasopressor dose series or the paired fluid actions and four-hour state transitions used in ICU-Sepsis. Nor does the one-row extract supply a propensity for selecting one of the 25 modeled intervention combinations. A retrospective mortality row might support an appropriately designed risk prediction question, with discharge/mortality fields excluded from baseline predictors, but cannot serve as off-policy treatment validation here. We count Dryad as one distinct, actually used sepsis data product; its 13,717 rows do not become 13,717 GEO GSM accessions or independent sequential trajectories. Exact aggregate counts, source hash and script are in results/sepsis_dryad_eicu.json and src/sepsis_dryad_eicu_audit.py. The policy-data gate remains zero.

25 Paired synthetic transition perturbations: a sensitivity check, not a new hospital

To see whether the seed-31 centroid-versus-one-hot value reversal is a fragile accident of one fixed transition table, we evaluated their already trained 150-epoch policies under twelve identically seeded perturbations of the released ICU-Sepsis transition matrix. For each state-action row, a Dirichlet draw retains precisely the source table’s nonzero next-state support and uses a pseudo concentration of ten. The value calculation holds all action maps fixed; no policy is retrained or selected on the perturbed values. The same sampled matrix is applied to one-hot, centroid and

exact original-model-optimal policies on each draw. This pairing removes Monte Carlo differences attributable solely to using different perturbations for the compared policies, but not uncertainty about actual ICU care. The concentration is invented and the source rows' empirical patient counts are unavailable.

For this synthetic assumption, mean values across the twelve draws were 0.858608 one-hot, 0.847394 centroid and 0.875441 for the original exact-optimal action map. The paired centroid-minus-one-hot model-value difference was negative in all twelve draws, averaging -0.011214 and spanning -0.022492 to -0.000917 . The source matrix itself gave 0.856932, 0.845072 and 0.875142 respectively. These twelve draws are not twelve independent patients, institutions or external transition systems; their spread is not a confidence interval. They resample a highly restricted transition support and omit changes in observation encoding, treatment propensities, case mix and outcome ascertainment. Furthermore, the seed-31 state split and architecture were inspected earlier, so repeating a convenient difference under hypothetical noise does not provide a preregistered generalization result. This within-project experiment supports only the narrow observation that the negative full-policy centroid contrast persists under this particular local, same-support stochastic stress test. Exact row draw settings and all paired outcomes are in `src/sepsis_policy_perturb_compare.py` and `results/sepsis_policy_perturb_compare.json`.

26 A second derived ICU data product: immune trajectories, not action histories

We separately fetched both public CSV files from the Figshare derived eICU and MIMIC-IV sepsis collection, https://figshare.com/articles/dataset/_b\Derived\ICU\Sepsis\Dataset\from_eICU_and_MIMIC-IV_Immune_Cell_Trajectories_SOFA_and_Outcomes_b_/31146940. This is a different published product from the static Dryad eICU extract above; because both derive in part from eICU, independent patient overlap cannot be ruled out. Figshare describes it as de-identified aggregated variables including neutrophil and lymphocyte trajectories, SOFA-related measures and outcomes. We use only aggregate analysis here, not the patient-level rows, and record the CSV SHA256 values in `results/sepsis_figshare_immune.json`.

The eICU file has 6,860 rows with 6,860 distinct patient-unit stays. The MIMIC-IV file has 2,423 rows with 2,423 distinct stays. Recorded neutrophil-to-lymphocyte ratios appear on days one through seven for 6,860, 5,915, 5,884, 4,774, 3,859, 3,145 and 2,536 eICU stays; MIMIC-IV counts are 2,423, 1,432, 1,377, 1,306, 1,165, 1,052 and 976. These are wide repeated-measure summaries, not seven separately accessioned cohorts. Missing later measurements need not be missing at random: discharge, death and clinical measurement decisions can govern observation. Using day-one ratio alone against the source's 28-day unit-death field produces a retrospective within-cohort rank AUC of 0.507 in eICU; against its distinct 28-day death field the MIMIC-IV rank AUC is 0.564. We do not interpret either descriptive number as a held-out prognostic result, and do not directly compare endpoints with different definitions.

The eICU file has a summary 'vasopressor_use' field, and MIMIC-IV has a 'vasopressin' field, but neither supplies the timed joint fluid/pressor interventions and ICU-Sepsis state transitions. The

two files can characterize immune trajectories and missingness, not fit or independently score our treatment policy. We count the Figshare collection as one actually used scientific data product, not two products per CSV and not 9,283 new accession-level policy datasets. The policy-trajectory and clinical-validation gates remain open.

27 A hospital sepsis process log: real time stamps but wrong interventions

The public Zenodo simplified sepsis event-log archive, <https://zenodo.org/records/3989590>, contains one excerpt from a real hospital information system and three explicitly synthetic amplified variants. We downloaded and hashed the archive, used only its ‘real.log’, and excluded ‘x10.log’, ‘x100.log’ and ‘x1000.log’ from counts. The original has 2,654 time-stamped activity events on 1,050 case IDs: 1,049 ER Sepsis Triage events, 823 IV Antibiotics events and 782 release events labeled A through E. The most common exact activity sequence is triage, antibiotic, Release A (589 cases). Across 820 cases with an eligible triage-to-antibiotic pair the descriptive median interval is 1.532 hours. These intervals may include workflow delays and the source supplies no adjudicated clinical outcomes; Release A through E cannot be casually recoded as mortality classes.

This is a distinct actually used public sepsis data product and a useful test of what the word “trajectory” does not guarantee. The only three fields in each event are case ID, activity label and Unix time stamp. IV Antibiotics records an event, not a dose, and the log has no four-hour physiologic state, paired IV fluid and vasopressor intervention, next-state measure or action propensity. Its cases cannot be matched to our ICU-Sepsis states or GEO sample accessions. Calling the three synthetic size amplifications extra patients would be false. The numerical audit is in `src/sepsis_zenodo_workflow.py` and `results/sepsis_zenodo_workflow.json`. We count one tool/data product, not 1,050 treatment-policy datasets; clinical and policy-trajectory gates remain open.

28 Randomized fluid-guidance arms do not yield dynamic-policy trajectories

A separate open Zenodo Stata dataset accompanies Sricharoenchai and colleagues’ single-blind randomized trial of dynamic inferior-vena-cava (IVC) versus static central-venous-pressure (CVP) guidance: <https://zenodo.org/records/10579408>. The associated paper is <https://pubmed.ncbi.nlm.nih.gov/articles/PMC11342037/>, DOI 10.12688/f1000research.147875.2. The authors report one Thai hospital, August 2016 to April 2020, APACHE II stratification and block randomization. The paper enrolled 124 participants (62 per arm); one in dynamic IVC withdrew, leaving 123 released rows, while one static CVP record lacks 30-day survival follow-up after transfer. This design is stronger evidence about the assigned guidance strategies than the observational vasopressor contrasts above, but the present audit is only a descriptive re-reading of the released data, not a new randomized-trial analysis.

The 58-field Stata file uses ‘intervention=0’ for static CVP (62 rows) and 1 for dynamic IVC (61 rows), as verified against its embedded variable label. Observed 30-day deaths are 28/61 (45.9%) in static CVP, with one missing status, and 21/61 (34.4%) in dynamic IVC. The source paper says that the primary mortality difference was not statistically significant ($p=0.196$), reporting relative risk 0.8 with 95% confidence interval 0.5–1.2 ($p=0.201$). It separately reports shorter shock duration and lower accumulated norepinephrine in the dynamic arm. In the released rows, median total 72-hour IV fluid is 8,081 mL in static and 8,542 mL in dynamic; 59/62 versus 52/61 have a recorded norepinephrine-use indicator. These numbers describe the published assignments and measured summaries, not an inference that one individual should receive a particular dose or that our controller improves mortality. The trial’s small sample and missing survival outcome are especially relevant to the uncertain mortality estimate.

Crucially, the ‘intervention’ column is an assigned monitoring/guidance strategy, not a repeated action chosen at every ICU decision step. Accumulated norepinephrine and 72-hour fluid fields contain no dose timestamps, patient observations at each decision or logged propensities. It cannot train or score a 25-action ICU-Sepsis policy. We count the Zenodo record as one distinct actually used sepsis scientific data product, not 123 treatment-policy trajectories or GEO accessions. Only aggregate calculations and source SHA256 appear in results/sepsis_zenodo_fluid_trial.json; the Stata patient rows remain outside Git. Clinical and policy-trajectory gates stay open.